Simplifying Statistics with SQL for Data Analysis

This is the second article in the series on data analysis. The previous article dealt with pivot tables. This article explores ways to find the measures of central tendency and the measures of variability using MySQL.

Structured Query Language (SQL) is a good starting point for high level data analysis. Many data analysis packages and languages have their own interface to read data from different SQL based database systems. In fact, any real-life data analysis starts from an RDBMS, and the basic analysis and report generation is done on the SQL platform of that RDBMS itself. A good preview of data within the RDBMS platform itself helps analysts to get a fast high level analysis in a different platform. Besides this, since SQL can combine multiple tables, the export of compact data from different tables in a rich data analysis platform like R, Python or SPSS can make for a powerful data analysis system.

Here, I will discuss only the basic descriptive statistical data analysis facilities of SQL. Some of the options are implicitly available and some can be derived from the primary facilities of the language.

Resources

The analysis has been done on the Northwind RDBMS available in the Google archive (*https://code.google.com/archive/p/northwindextended/downloads*). Readers may download the data and can experiment with it in their own way.

Descriptive statistics

Data analysis is slightly different from general SQL based queries. While the basic purpose of the SQL query is to prepare a report, the purpose of data analysis is to analyse the report and come to a conclusion. As data analysis requires proper organisation of data in a particular context, an SQL query is an essential prerequisite. Data analysis is a vast and complex subject, and uses different supervised and unsupervised techniques. In this article, I will only cover the descriptive statistics to analyse data, in order to get some basic trends inherent to the distribution, so that an analyst can derive meaningful conclusions from the data and initiate a high level analysis of the data distributed over a set of tables.

For numerical data, descriptive statistics, in general, measures the central tendency (mean, median, mode) and the level of variability of the data. These measurements are a good starting point for data exploration and often lead to new questions that fuel further higher level analysis.

Most of the required analysis of data is the aggregation of data, as a whole or in groups. This is needed in order to have an overall view of the records. Aggregate functions allow us to easily produce summarised data from our database. With aggregation functions like *MAX(), MIN(), AVG(),SUM()* and *COUNT()* one can easily produce reports from database tables. Aggregate functions are all about performing the following:

- Calculations on multiple rows
- Calculations on a single column of a table
- Calculation to get a single value of a parameter

For instance, from the Northwind database, the management may require the following analysis reports: